

SUMMARY

This video elucidates key insights from the instructional session on General Aptitude for GATE, specifically focusing on the Number System. It introduces structured preparation based on mark distribution and problem frequency. Core concepts include classification of numbers, factorial properties, prime factor counting, and trailing zero determination. Special emphasis is given to the “GC Trick” (Good Conceptual Trick) for faster calculations and the “Chotu is the Hero” principle, which states that the limiting (minimum) factor determines the final outcome in pairing-based problems.

1. GENERAL APTITUDE COURSE STRUCTURE AND MARK DISTRIBUTION

Understanding mark distribution helps prioritize preparation effectively.

Category: Quantitative Aptitude

Estimated Marks: 7–8

Key Topics: Number System, Numerical Computation (Profit/Loss, Speed-Time-Distance), Probability, Permutations & Combinations (PnC), Data Interpretation (DI), Geometry, Statistics

Category: Analytical Aptitude

Estimated Marks: 3–4

Key Topics: Logical reasoning and pattern identification

Category: Verbal Aptitude

Estimated Marks: ~3

Key Topics: Language and comprehension

Category: Spatial Aptitude

Estimated Marks: 1–2

Key Topics: Visualization and spatial reasoning

Critical Takeaways:

Data Interpretation – Guaranteed 2-mark question

Geometry – Guaranteed 2-mark question

Numerical Computation – Guaranteed 2-mark question

Number System/Missing Numbers – Guaranteed 2-mark question

2. FOUNDATIONAL NUMBER SYSTEM CONCEPTS

The number system spans from negative infinity to positive infinity on the real number line.

2.1 CLASSIFICATION OF NUMBERS

Natural (Counting) Numbers: Start from 1

Whole Numbers: Start from 0

Even Numbers: Divisible by 2

Odd Numbers: Not divisible by 2

Prime Numbers: Divisible only by 1 and themselves; the smallest even prime is 2. Prime numbers are considered the “hero” in number system problems.

Composite Numbers: Formed by multiplication of prime numbers

Complex Numbers: Result from the square root of negative numbers; excluded from GA syllabus

2.2 FACTORIALS (FACTORS IN SERIAL)

Factorial ($n!$) is the product of all integers from 1 to n .

$$0! = 1$$

$$1! = 1$$

$$5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$$

3. ADVANCED FACTORIAL ANALYSIS: FINDING PRIME FACTORS

A common GA question involves determining how many times a prime number appears in $n!$.

3.1 CONVENTIONAL FORMULA

Count of prime r in $n!$ =

$$\lfloor n/r \rfloor + \lfloor n/r^2 \rfloor + \lfloor n/r^3 \rfloor + \dots$$

Continue until the denominator exceeds the numerator.

3.2 THE “GC TRICK” FOR RAPID CALCULATION

Instead of calculating powers:

1. Divide n by the prime r .
2. Take the integer quotient and divide again by r .
3. Repeat until the quotient becomes less than r .
4. Add all quotients.

Example: Number of 2s in $100!$

$$100/2 = 50$$

$$50/2 = 25$$

$$25/2 = 12$$

$$12/2 = 6$$

$$6/2 = 3$$

$$3/2 = 1$$

$$\text{Total} = 50 + 25 + 12 + 6 + 3 + 1 = 97$$

4. DETERMINING NUMBER OF TRAILING ZEROS

Trailing zeros are formed by pairing 2 and 5 (since $2 \times 5 = 10$).

4.1 THE “CHOTU IS THE HERO” PRINCIPLE

The number of trailing zeros equals the minimum count between 2s and 5s.
Since 2s are usually more abundant in factorials, the number of 5s determines the zeros.

4.2 APPLICATION RULES

In Factorials ($n!$):

Number of Zeros = Number of 5s

Example: In $100!$

$$100/5 = 20$$

$$20/5 = 4$$

Total = 24 zeros

In General Expressions:

Count total 2s and 5s across all factors and take the minimum.

In Powers (X^P):

Total Zeros = (Zeros in X) $\times$ P

In Sums ($A + B + C$):

The number of trailing zeros is determined by the term with the fewest trailing zeros.

Example: $100 + 10 = 110$

The result has 1 trailing zero because 10 has only 1 trailing zero.

5. KEY PROBLEM-SOLVING INSIGHTS

5.1 COMPOSITE FACTORS

To determine the occurrence of a composite number (e.g., 6) in $n!$:

Break it into prime factors.

$$6 = 2 \times 3$$

Number of 6s in $100!$ = min(count of 2s, count of 3s)

5.2 HIDDEN FACTORS

Some expressions hide factors like 5 within algebraic simplifications.

Example:

$$3^{98} - 3^{97} - 3^{96}$$

Factor out 3^{96} :

$$3^{96}(3^2 - 3^1 - 3^0)$$

$$= 3^{96}(9 - 3 - 1)$$

$$= 3^{96}(5)$$

This reveals a hidden factor of 5, which can pair with a 2 to form a trailing zero.

5.3 COMPLEX FACTORIAL PRODUCTS

Example:

$$6!^{(6!)} \times 8!^{(8!)} \times 10!^{(10!)}$$

6! has 1 trailing zero $\rightarrow$ contributes $1 \times 6!$ zeros

8! has 1 trailing zero $\rightarrow$ contributes $1 \times 8!$ zeros

10! has 2 trailing zeros $\rightarrow$ contributes $2 \times 10!$ zeros

$$\text{Total Zeros} = 6! + 8! + 2(10!)$$

